Practical 4 – Sales Data Clustering

Exactly 60 Viva Questions & Answers | K-Means, Elbow Method, Hierarchical Clustering and PCA

Scope: Your original code implements K-Means and the Elbow Method. Hierarchical Clustering and PCA are included as theory topics from your practical manual.

A. Clustering and Dataset Basics

Q1. What is clustering?
An unsupervised learning technique that groups data points based on similarity. Points in one cluster should be more similar to each other than to points in other clusters.

Q2. Why is K-Means unsupervised?
It discovers groups from input features without being given predefined class labels or a target output.

Q3. What is the objective of this practical?
To group similar sales records and estimate a suitable number of groups using the Elbow Method.

Q4. What is a feature?
An input variable used by an algorithm. This practical selects QUANTITYORDERED, PRICEEACH, SALES and MSRP.

Q5. What do the four selected features mean?
QUANTITYORDERED is item quantity; PRICEEACH is unit price; SALES is sales amount; MSRP is manufacturer's suggested retail price.

Q6. Why is there no target variable Y?
This is unsupervised learning. The algorithm groups observations using X without predicting a known target.

Q7. What is a centroid?
The mean position of all points assigned to a cluster. It represents the centre of that cluster in feature space.

Q8. What does K mean in K-Means?
K is the requested number of clusters. For example, K = 3 asks the algorithm to create three groups.

B. K-Means Algorithm and Mathematics

Q9. How does K-Means work?
Choose K initial centroids, assign each point to its nearest centroid, recalculate each centroid as the mean of its assigned points, and repeat until assignments or centroids stabilize.

Q10. Why is it called centroid-based?
Each cluster is represented by a centroid, and each observation is assigned to the nearest centroid.

Q11. How is a new centroid calculated?
For each feature, calculate the arithmetic mean of that feature among all points assigned to the cluster. The means form the new centroid vector.

Q12. Which distance measure is commonly used here?
Euclidean distance. Scikit-learn's standard KMeans uses squared Euclidean distances for its inertia objective.

Q13. What is the Euclidean distance formula?
d(x,y) = sqrt(sum over i of (x_i - y_i)^2). It measures straight-line distance between two points in feature space.

Q14. What is the K-Means objective function?
Minimize WCSS = sum from j=1 to K, then for each x_i in cluster C_j, of ||x_i - mu_j||^2.

Q15. Explain the symbols in WCSS.
K is number of clusters; C_j is cluster j; x_i is a data point; mu_j is the centroid of cluster j; the squared norm is squared distance to that centroid.

Q16. Why are centroids recalculated?
The mean represents the current cluster members. Updating it can reduce the total squared distance between members and their centroid.

Q17. What are K-Means limitations?
It requires K, is sensitive to outliers and initialization, and works best for compact, roughly spherical clusters. It can struggle with irregular shapes or varying densities.

Q18. Can K-Means produce different results on different runs?
Yes. Different initial centroids can lead to different local solutions. A fixed random seed makes runs reproducible; multiple initializations help find a better solution.

C. Preprocessing and Your Python Code

Q19. Why is Pandas imported?
Pandas loads and manipulates tabular data. read_csv loads the CSV file into a DataFrame.

Q20. What does df.shape return?
A tuple containing the number of rows and columns, in the form (rows, columns).

Q21. Why select only numerical columns?
K-Means calculates numerical distances. The selected numeric sales features can be used directly in those calculations.

Q22. Why handle missing values before K-Means?
Missing numerical values prevent reliable distance calculations. Imputation replaces missing entries with usable values.

Q23. What does SimpleImputer(strategy='median') do?
It replaces missing values in each feature with that feature's median.

Q24. Why use median instead of mean?
Median is less affected by extreme values and can represent a typical value better in skewed data.

Q25. Explain fit(), transform() and fit_transform().
fit() learns preprocessing information; transform() applies it; fit_transform() learns and applies the operation in one call.

Q26. Why is feature scaling needed?
K-Means is distance-based. A feature with a much larger numerical range could dominate distances, so scaling makes features more comparable.

Q27. What does StandardScaler do?
It standardizes each feature to approximately zero mean and unit standard deviation, based on the data used to fit it.

Q28. Give the StandardScaler formula and explain it.
z = (x - mu) / sigma. x is the original value, mu is the feature mean, sigma is its standard deviation, and z is the standardized value.

Q29. Normalization vs standardization?
Min-max normalization commonly maps values to a fixed range such as 0–1. Standardization centres values around zero and scales by standard deviation; it does not guarantee a fixed range.

Q30. Why use random_state=42?
It fixes the random seed so centroid initialization and results can be reproduced under the same conditions. 42 has no special mathematical meaning.

Q31. What does n_init=10 mean?
K-Means is run with 10 different centroid initializations, and the run with the lowest inertia is selected.

D. Elbow Method and Model Output

Q32. What is the Elbow Method?
A heuristic for estimating K. Run K-Means for several K values, record WCSS, plot WCSS against K, and choose the bend where improvement begins to diminish.

Q33. What does WCSS stand for?
Within-Cluster Sum of Squares: the sum of squared distances from each point to its assigned cluster centroid.

Q34. What is inertia_ in Scikit-learn?
The sum of squared distances of samples to their closest cluster centre. In standard K-Means, this is the WCSS objective value.

Q35. Why test K from 1 to 10?
It compares clustering compactness over a chosen range of candidate cluster counts. The range is a programmer's choice, not a fixed rule.

Q36. Why does WCSS generally decrease as K increases?
More clusters allow points to be assigned to closer centroids. The minimum achievable WCSS cannot increase when an extra cluster is allowed.

Q37. Why not choose the largest K?
A very large K can over-segment the data and create unnecessary complexity. The method seeks a balance between compactness and simplicity.

Q38. What does the elbow point represent?
The approximate K after which adding clusters produces a much smaller reduction in WCSS. It is a visual heuristic and may not be clear in every dataset.

Q39. What are the axes of the Elbow graph?
X-axis is number of clusters K; Y-axis is WCSS or inertia.

Q40. What does wcss.append(kmeans.inertia_) do?
It stores the current model's inertia in the WCSS list so it can be plotted against the corresponding K.

Q41. fit() vs fit_predict()?
fit() learns cluster centres. fit_predict() learns the centres and returns a cluster label for every observation.

Q42. What do labels 0, 1 and 2 mean?
They are arbitrary identifiers for groups. A higher label does not mean a better, larger or more important cluster.

Q43. Why add df['Cluster'] = clusters?
It attaches the assigned cluster identifier to each original sales record for inspection and analysis.

Q44. What does value_counts() show for Cluster?
It counts how many records have each cluster label, showing the size of each cluster.

Q45. Why use inverse_transform() on cluster_centers_?
Centres are in standardized units. inverse_transform converts them to original units, making quantity, price, sales and MSRP easier to interpret.

Q46. Does the scatter plot show all four clustering features?
No. Clustering uses all four scaled features, but the 2D scatter plot displays only the first feature, QUANTITYORDERED, and the third, SALES.

E. Hierarchical Clustering and PCA Theory

Q47. What is Hierarchical Clustering?
An unsupervised method that builds a hierarchy of nested clusters, commonly represented using a dendrogram.

Q48. What is agglomerative clustering?
The bottom-up approach: each observation starts as its own cluster, and the closest clusters are repeatedly merged.

Q49. What is divisive clustering?
The top-down approach: all observations start in one cluster, which is repeatedly split into smaller groups.

Q50. What is a dendrogram?
A tree-shaped diagram showing the order and distances at which clusters are merged. Cutting it at a chosen height produces groups.

Q51. What is single linkage?
The distance between two clusters is the minimum distance between any point in one cluster and any point in the other.

Q52. What is complete linkage?
The distance between two clusters is the maximum pairwise distance between points in the two clusters.

Q53. What is average linkage?
The distance between two clusters is the average of distances between all pairs of points, one from each cluster.

Q54. What is centroid linkage?
The distance between two clusters is calculated using the distance between their centroids.

Q55. How does K-Means differ from Hierarchical Clustering?
K-Means partitions data around K centroids and requires K in advance. Hierarchical clustering builds a merge/split hierarchy and uses a dendrogram; K is not needed to build the hierarchy.

Q56. What is PCA?
Principal Component Analysis is a dimensionality-reduction technique that transforms correlated features into orthogonal principal components retaining as much variance as possible.

Q57. Why is PCA used?
It reduces dimensions, simplifies visualization, reduces redundancy among correlated features and can lower computational

cost.

Q58. What are eigenvalues and eigenvectors in PCA?
Eigenvectors give principal-component directions; corresponding eigenvalues indicate how much variance is captured along those directions.

Q59. What is the role of the covariance matrix in PCA?
It summarizes how pairs of features vary together. PCA finds its eigenvectors and eigenvalues to identify principal directions and their associated variance.

Q60. Is PCA clustering, and is it in your code?
No. PCA reduces dimensions and does not assign cluster labels. It is theory in the manual but is not implemented in the original K-Means code.

Formula Sheet

Euclidean distance
d(x,y) = sqrt(sum(i=1 to n) (x_i - y_i)^2)
Straight-line distance between two points in n-dimensional feature space.

Cluster centroid
mu_j = (1 / |C_j|) * sum(x_i in C_j) x_i
Mean vector of all points assigned to cluster j.

WCSS / K-Means objective
WCSS = sum(j=1 to K) sum(x_i in C_j) ||x_i - mu_j||^2
Total squared distance from each point to its assigned centroid.

Standardization
z = (x - mu) / sigma
x = original value; mu = feature mean; sigma = standard deviation; z = standardized value.

PCA covariance matrix
C = (Z^T Z) / (n - 1)
For centred/standardized Z; Z^T is transpose and n is number of observations.

Important Code Functions

Code / Attribute	Purpose
pd.read_csv()	Loads the CSV dataset.
df.shape	Returns row and column counts.
SimpleImputer(strategy='median')	Fills missing values using feature medians.
StandardScaler()	Standardizes numerical features.
KMeans(...)	Creates a K-Means model.
kmeans.fit(X_scaled)	Fits the model to scaled data.
kmeans.inertia_	Returns WCSS / squared distances to nearest centres.
kmeans.fit_predict(X_scaled)	Fits K-Means and returns cluster labels.
kmeans.cluster_centers_	Returns learned centres in scaled feature space.
scaler.inverse_transform()	Converts scaled centroids to original units.
value_counts()	Counts observations in each cluster.
plt.scatter()	Displays a two-dimensional scatter plot.

Last-Minute Recall

• K-Means is unsupervised and centroid-based; K is the number of clusters.

• Your code selects QUANTITYORDERED, PRICEEACH, SALES and MSRP.

• Median imputation handles missing values; StandardScaler prevents large scales dominating distance.

• The Elbow Method tests K = 1 to 10, records inertia_ (WCSS), and plots K against WCSS.

• The elbow is where additional clusters give diminishing improvement; it is an estimate, not a guarantee.

• n_init=10 tries ten initializations; random_state=42 supports reproducibility.

• fit_predict() returns labels; labels are identifiers, not rankings.

• inverse_transform() converts scaled centroids back to original units.

• The model clusters using four features; the final scatter plot displays Quantity Ordered and Sales only.

• Your original code implements K-Means and Elbow Method, not Hierarchical Clustering or PCA.

• Agglomerative = bottom-up merging; divisive = top-down splitting; dendrogram = tree representation.

• PCA reduces dimensions; eigenvectors give component directions and eigenvalues indicate captured variance.

YouTube Video List

YouTube Video List

• **K-Means Clustering – StatQuest**
https://www.youtube.com/watch?v=4b5d3muPQmA

• **Hierarchical Clustering – StatQuest**
https://www.youtube.com/results?search_query=StatQuest+Hierarchical+Clustering

• **Principal Component Analysis (PCA) – StatQuest**
https://www.youtube.com/results?search_query=StatQuest+Principal+Component+Analysis+PCA

• **K-Means Clustering with Python – Codebasics**
https://www.youtube.com/watch?v=EItlUEPCIzM